Alaa Faraj

+970594528583 | alaaemadf4@gmail.com | Ramallah, Palestine | linkedin.com/in/alaa-faraj-2b26a2209 | github.com/alaax159

Education

Birzeit University Sep 2022 – Present
Bachelor of Computer Engineering

Skills

Languages: Python Programming, C, Java, JavaScript, TypeScript, SQL
AI / ML: Machine Learning, Deep Learning, Data Preprocessing, Data Augmentation, Hyperparameter Tuning, Model Evaluation & Optimization
Frameworks / Libraries: TensorFlow, Keras, HuggingFace Transformers, scikit-learn, FastAPI
Databases: MySQL, SQL Queries, Database Design

Projects

OffPI – Offline Emergency Communication System | Devpost Feb 2026

- Built a fully offline emergency communication platform at a 48-hour hackathon, combining an iOS app with Raspberry Pi edge nodes and LoRa long-range radio.
- Implemented P2P mesh networking, store-and-forward buffering, and offline AI inference for emergency classification.

AI Legal Aid for Palestine | GitHub 2026

- Built a voice-based AI legal assistant helping Palestinian citizens understand their rights under the Palestinian Basic Law, supporting both Arabic and English.
- Implemented a full RAG pipeline using ChromaDB and multilingual sentence transformers; integrated Whisper for Arabic speech-to-text, FastAPI backend, and PDF form auto-fill for legal documents.
- **Technologies:** Python, FastAPI, ChromaDB, HuggingFace Transformers, Whisper, SaulLM-7B, QLoRA, reportlab

Image Classifier with TensorFlow (Nanodegree) | GitHub Jun 2025 – Aug 2025

- Designed and trained CNN models using TensorFlow/Keras, improving validation accuracy through data preprocessing, data augmentation, and hyperparameter tuning as part of the Udacity AI Developer Nanodegree.
- Built a full ML pipeline (train/val/test split) and developed Python programming inference scripts for predicting new images, reducing inference time by optimizing model architecture.
- **Technologies:** Python, TensorFlow, Keras, NumPy, Matplotlib

Time-of-Day Classification – Multimodal ML | GitHub Jan 2026

- Built a multimodal time-of-day classifier (morning/afternoon/evening) comparing three approaches: KNN baseline on handcrafted color features, CNN using EfficientNetV2-L transfer learning (87.4% test accuracy), and RoBERTa Transformer on text descriptions (76.4% test accuracy).
- Applied transfer learning, partial fine-tuning, data augmentation, and early stopping; conducted EDA on imbalanced dataset and evaluated models using accuracy, F1-score, and confusion matrices, improving overall F1-score by 12% over the KNN baseline.
- **Technologies:** Python, TensorFlow, Keras, HuggingFace Transformers, RoBERTa, EfficientNetV2, scikit-learn

Jewelry Shop Database System | GitHub Jan 2023 – May 2023

- Designed a complete relational database with 8+ tables (ER diagram, normalization, referential integrity) and developed SQL queries for CRUD operations and business reporting; simulated real business scenarios.
- **Technologies:** MySQL, SQL Queries, Database Design (ERD, Normalization)

Certifications

Intro to Machine Learning with TensorFlow (Nanodegree) | Udacity – Palestine Launchpad Mar 2026

- Credential: udacity.com/certificate/e/6b3c012c-0060-11f1-b716-f7f806e7cfc4

AI Programming with Python & TensorFlow (Nanodegree) | Udacity Jul 2025

- Credential ID: 90-edf2-11ef-a0d3-83376eefa3be

Languages

Arabic: Native **English:** B2 Upper Intermediate (EFSET Certified)